End-Semester Project Report:

Multi-Class Glioma Brain Tumour Segmentation Using XGBoost with MRF Post-Processing on the BraTS 2024 Dataset

Shashwat Gohel
Kavish Shah
Aarav Jhaveri
Yash Shah

School of Engineering and Applied Sciences, Ahmedabad University
Enrollment: AU2340235, AU2340244, AU2340249, AU2340078 Professor: Dr. Mehul Raval

Teaching Assistant: Shalvi Modi

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Shashwat Gohel, Kavish Shah, Aarav Jhaveri, Yash Shah
Ahmedabad University

***Abstract*—The following study gives the results of the multi-class segmentation algorithm for brain tumors using machine learning. The method used here is applied to the MP-MRI images available in the BraTS 2024 dataset, and it includes five major phases: (i) z-score normalization of brain-masked data, (ii) creation of a 109-dimensional feature vector per voxel (multi-scale Gaussian, Gabor, gradient, Laplacian, DoG, entropy estimation, and spatial feature maps), (iii) building the XGBoost multi-class classification model (`multi:softprob`) trained with 4.05 million voxels obtained from 1,214 samples, (iv) mean-field approximation of Markov Random Fields (MRFs) ($\beta = 0.5$), and (v) probability boosting with thresholding optimization.**

The Dice score for the proposed framework is calculated using 136 test cases and found to be 0.8279 for WT, 0.6849 for TC, and 0.6787 for ET, along with their IoU values of 0.7248, 0.5683, and 0.5628, respectively. The maximum difference between the training and testing Dice value is 0.0438, which reflects negligible overfitting in the network. This method outperforms the previous Phase 1 binary WT (Dice: 0.7508) by 10.3% in terms of accuracy while providing complete segmentation of sub-regions.

***Index Terms*—glioma segmentation, BraTS 2024, XGBoost, Markov Random Field, multi-class segmentation, feature engineering, Gabor filters, MRF post-processing, medical image analysis, traditional machine learning**

I. Introduction

Brain tumours, particularly the more aggressive tumours such as Glioblastoma Multiforme (GBM), still form part of the most lethal types of cancers. It is essential to determine the location of the tumour based on multimodal Magnetic Resonance Imaging (MRI) scans. For brain tumour segmentation, there are three key areas highlighted in the BraTS challenge [1] as follows:

- **Whole Tumour (WT)**: union of NCR + Edema + ET, labels $\{1, 2, 3\}$.
- **Tumour Core (TC)**: NCR + ET, labels $\{1, 3\}$.
- **Enhancing Tumour (ET)**: actively growing region for enhancement, labeling $\{3\}$.

The BraTS 2024 data includes four MR images per subject: T1c, T1n, T2f, T2w, and they are $182\times218\times182$ pixel images after undergoing skull stripping and image registration.

Without the use of deep learning, this project has been able to demonstrate that by designing voxel-based features along with gradient boosting techniques, one can obtain similar results. The total project time taken was eight weeks:

- **Phase 1 (Weeks 1–5):** Baseline: Binary WT with 5 features Random Forest; WT Dice Score = 0.7508.
- **Phase 2 (Weeks 6–8):** Multi-class segmentation with XGBoost utilizing 109 features, then processed with MRF technique, and threshold optimization for each class.

II. Dataset and Preprocessing

A. Dataset Statistics

The splitting strategy for BraTS 2024 dataset is: Train-1,012 samples, Validation-202 samples, Test-136 samples unseen before. Relabel ET from BraTS label 4 to label 3. The internal scheme is: 0 (BG), 1 (NCR), 2 (Edema), 3 (ET).

B. Brain Mask Construction

$$M_{\text{brain}} = \bigcup_{m\in\{\text{T1c,T1n,T2f,T2w}\}} \mathbf{1}[V_m > 0] \tag{1}$$

Mask average has 1.0–1.8 million voxels per patient.

C. Z-Score Normalisation

$$\hat{V}_m(x) = \begin{cases} \dfrac{V_m(x) - \mu_m}{\sigma_m + \varepsilon} & x \in M_{\text{brain}} \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

with $\varepsilon = 10^{-8}$; μ_m, σ_m computed over brain voxels only.

D. Class Imbalance

This dataset is imbalanced on an extreme level. Distribution of sampling is provided in Table I.

TABLE I: Label Distribution in the Sampled Training + Validation Set

ID	Label	Sampled Voxels	Fraction
0	Background	3,919,389	96.7%
1	NCR/NET	7,389	0.2%
2	Edema	52,738	1.3%
3	ET	73,826	1.8%
Total		**4,053,342**	**100%**

1:49 tumour-to-background ratio; NCR 4×, ET 3× oversampled; capped at 4M rows.

III. Feature Engineering

A. Overview

The Brain voxel is represented by the 109 dimensions of feature vector as indicated in Table II.

TABLE II: Complete Feature Set Composition (109 Features Per Voxel)

Feature Group	Per Mod.	Total
Raw intensity	1	4
Multi-scale mean (win 3,5,9,15)	4	16
Multi-scale std (win 3,5,9,15)	4	16
Gradient magnitude (3-D Sobel)	1	4
Laplacian	1	4
Multi-scale Gaussians (σ=1, 2, 3)	3	12
$\mathrm{DoG}_{1\text{-}2}$, $\mathrm{DoG}_{2\text{-}3}$	2	8
Gabor ($0°, 45°, 90°, 135°$)	4	16
Asymmetry (L–R flip diff.)	1	4
Entropy proxy ($\sigma_{\text{loc}}/\mu_{\text{loc}}$)	1	4
Distance to brain edge (EDT)	1	4
Subtotal per-modality (×4)	*23*	*92*
Cross-modal ratios, products, diff.	–	13
Spatial coords (x, y, z, dist-centroid)	–	4
Total		**109**

B. Per-Modality Features

Multi-scale statistics. Local mean and standard deviation via box-filtering:

$$\sigma^2_{\text{local}}(x) = \overline{V^2}(x) - \left[\overline{V}(x)\right]^2 \tag{3}$$

Gradient magnitude.

$$|\nabla V|(x) = \sqrt{G_x^2 + G_y^2 + G_z^2} \tag{4}$$

Difference-of-Gaussians.

$$\mathrm{DoG}_{1,2} = G_{\sigma=1}(V) - G_{\sigma=2}(V) \tag{5}$$

$$\mathrm{DoG}_{2,3} = G_{\sigma=2}(V) - G_{\sigma=3}(V) \tag{6}$$

Gabor texture. 4 responses oriented at the frequency of 0.2; kernel is pseudo-3D convolution: $k^{3D}(u, v, w) = k^{2D}(u, v)$.

Asymmetry.

$$A_m(x, y, z) = V_m(x, y, z) - V_m(H - x, y, z) \tag{7}$$

Entropy proxy.

$$H_{\text{proxy}}(x) = \frac{\sigma_{\text{local}}(V, s = 5)}{\mu_{\text{local}}(|V|, s = 5) + 10^{-6}} \tag{8}$$

Distance to brain edge. Euclidean distance map of the brain mask range normalized to $[0, 1]$.

C. Cross-Modal and Spatial Features

Cross-modal features (13 ratios/differences between T1c and T1n, products, differences between T2f and T2w), four spatial locations.

IV. Data Sampling and Class Balancing

All 1,214 training+validation patients are combined. A ratio of 1:49 is applied between tumour and background data; the size limit is 4,000,000 entries. Oversampling factor is 4× for NCR and 3× for ET samples. Weights for each sample during training:

$$w_i = \begin{cases} 1.0 & \text{Background} \\ 8.0 & \text{NCR (hardest)} \\ 2.0 & \text{Edema} \\ 6.0 & \text{ET} \end{cases} \tag{9}$$

V. XGBoost Classifier

A. Hyperparameters

A table III presents the final settings for the XGBoost model to be used in all experiments.

TABLE III: XGBoost Classifier Hyperparameters

Hyperparameter	Value	Rationale
Objective	`multi:softprob`	Per-class probabilities
Num classes	4	BG, NCR, Edema, ET
Num estimators	500	Early stopping on mlogloss
Max depth	9	Complex voxel interactions
Learning rate	0.07	Balanced convergence
Subsample	0.85	Row subsampling
Colsample bytree	0.75	Feature subsampling
Colsample bylevel	0.75	Per-level subsampling
Min child weight	5	Prevents tiny-cluster splits
γ	0.1	Conservative split
α (L1)	0.1	Sparse feature weights
λ (L2)	1.0	Leaf weight shrinkage
Early stopping	30 rounds	Patience on mlogloss
Tree method	`hist`	Efficient on large data

B. Training Protocol

Dataset is divided as 93.6% / 6.4% training/evaluation where training has (3,753,342) examples and evaluation contains (300,000) examples. Training reached convergence at iteration 499 where validation `mlogloss` = **0.02874**. Table IV shows convergence graph.

TABLE IV: XGBoost Training Convergence (mlogloss on 300K Evaluation Slice)

Iter	mlogloss	Iter	mlogloss
0	0.21167	300	0.03189
50	0.05214	350	0.03081
100	0.04168	400	0.02996
150	0.03761	450	0.02927
200	0.03503	**499**	**0.02874**
250	0.03328		

C. Internal Evaluation Results

Table V reports class-wise precision, recall, and F1 for the 300K validation split.

TABLE V: Classification Report on 300K Internal Evaluation Slice

Class	Prec.	Rec.	F1	Support
Background	1.00	1.00	1.00	290,018
NCR/NET	0.78	0.69	0.73	515
Edema	0.84	0.77	0.80	3,973
ET	0.83	0.89	0.86	5,494
Accuracy			0.99	300,000
Macro avg	0.86	0.84	0.85	–

Region Dice on evaluation slice: WT = 0.8731, TC = 0.8690, ET = 0.8582. The final model is 17.3 MB on disk.

VI. MRF Post-Processing

A. Mean-Field Variational Inference

$$\log Q^{(t+1)}(x_i{=}c) = \log P_{\text{unary}}(x_i{=}c) - \beta \sum_{c'} \phi_{c,c'}\, \bar{Q}^{(t)}_{c'}(i) \tag{10}$$

with $\beta = 0.5$; $\bar{Q}^{(t)}_{c'}$ is spatially smoothed via Gaussian ($\sigma = 1$); 5 iterations.

B. Label Compatibility Matrix

Rows/cols: [BG, NCR, Edema, ET]:

$$\phi = \begin{pmatrix} 0.0 & -1.0 & -0.5 & -1.0 \\ -1.0 & 0.0 & -0.8 & 0.0 \\ -0.5 & -0.8 & 0.0 & -0.8 \\ -1.0 & 0.0 & -0.8 & 0.0 \end{pmatrix} \tag{11}$$

Adjacencies deemed biologically unlikely (BG–ET: -1.0) are heavily weighted; the NCR-ET adjacency (0.0) is allowed.

C. Small Component Removal

Blob components smaller than 200 voxels are excluded. Mean number of blobs excluded per test patient: **429.6**.

VII. Threshold and Boost Tuning

Tuning is only done using 202 validation samples in a two-pass optimization process.

Pass A: Sweeps WT threshold $\in \{0.15, 0.25, 0.35, 0.45, 0.55\}$ and NCR boost $\in \{1.5, 2.0, 2.5, 3.0, 3.5\}$; ET boost fixed at 2.0.

Pass B: Fine-tunes WT threshold near Pass A optimum at $\{0.47, 0.52, 0.57\}$; sweeps ET boost $\in \{1.5, 2.0, 2.5, 3.0\}$; NCR boost fixed.

TABLE VI: Top-5 Configurations from Pass A Tuning (202 Validation Patients)

WT thr	NCR b	ET b	WT	TC	ET	Mean
0.55	1.5	2.0	0.8329	0.6798	0.6745	0.7291
0.25	1.5	2.0	0.8329	0.6756	0.6702	0.7262
0.45	1.5	2.0	0.8329	0.6756	0.6702	0.7262
0.35	1.5	2.0	0.8329	0.6756	0.6702	0.7262
0.15	1.5	2.0	0.8329	0.6756	0.6702	0.7262

TABLE VII: Top-5 Configurations from Pass B Fine-Tuning (202 Validation Patients)

WT thr	NCR b	ET b	WT	TC	ET
0.52	**1.5**	**1.5**	**0.8360**	**0.6858**	**0.6816**
0.47	1.5	1.5	0.8359	0.6843	0.6800
0.57	1.5	1.5	0.8342	0.6881	0.6841
0.52	1.5	2.0	0.8332	0.6773	0.6719
0.47	1.5	2.0	0.8329	0.6756	0.6702

Optimal (bold): **WT threshold = 0.52, NCR boost = 1.5, ET boost = 1.5**; boost vector $[1.0,\ 1.5,\ 1.0,\ 1.5]$.

VIII. Quantitative Results

A. Test Set Performance (136 Patients)

Table VIII shows the Dice and IoU scores obtained for the 136 test samples that are not seen during training.

TABLE VIII: Test Set Evaluation Results — 136 Unseen Patients

Region	Mean Dice	Std	Median	Mean IoU
WT	0.8279	0.1420	0.8612	0.7248
TC	0.6849	0.2579	0.7986	0.5683
ET	0.6787	0.2641	0.7893	0.5628

Min/Max for all regions: 0.0000 / $\geq$0.96.

Patients exceeding Dice $\geq$ 0.70: WT = 123/136 (90.4%), TC = 91/136 (66.9%), ET = 90/136 (66.2%).

B. Training Set Performance (1,012 Patients)

Table IX shows the same measures calculated from the entire training dataset of 1,012 samples.

TABLE IX: Training Set Evaluation Results — 1,012 Patients

Region	Mean Dice	Std	Median	Mean IoU
WT	0.8347	0.1209	0.8669	0.7313
TC	0.7267	0.2298	0.8124	0.6120
ET	0.7219	0.2324	0.8105	0.6066

C. Overfitting Analysis

Table X compares the train and test scores; an overfitting warning would be issued if the difference exceeds 0.05.

TABLE X: Train vs. Test Comparison (flag threshold: gap $>$ 0.05)

Metric	Train	Test	Gap	Overfit?
WT Dice	0.8347	0.8279	+0.0068	**No**
TC Dice	0.7267	0.6849	+0.0418	**No**
ET Dice	0.7219	0.6787	+0.0432	**No**
WT IoU	0.7313	0.7248	+0.0065	**No**
TC IoU	0.6120	0.5683	+0.0437	**No**
ET IoU	0.6066	0.5628	+0.0438	**No**

D. Phase 1 vs. Phase 2

Table XI provides a summary of the improvement made over the binary classifier baseline to the full multi-class system.

TABLE XI: Phase 1 Binary RF vs. Phase 2 Multi-Class XGBoost

Metric	Phase 1	Phase 2	Δ
WT Dice	0.7508	0.8279	**+10.3%**
WT IoU	0.6269	0.7248	**+15.6%**
TC Dice	N/A	0.6849	New capability
ET Dice	N/A	0.6787	New capability

Phase 1: 5-feature Random Forest. Phase 2: 109-feature XGBoost + MRF.

IX. Qualitative Results and Visualisations

Predictions based on the optimized setting (WT threshold = 0.52, boost vector $[1.0, 1.5, 1.0, 1.5]$) are colour coded here: **Red** = NCR, **Yellow** = Edema, **Cyan** = ET.

A. High-Performing Test Case

Patient BraTS-GLI-03056-101: WT = 0.8599, TC = 0.7888, ET = 0.7945. Boundary of all three regions predicted accurately.

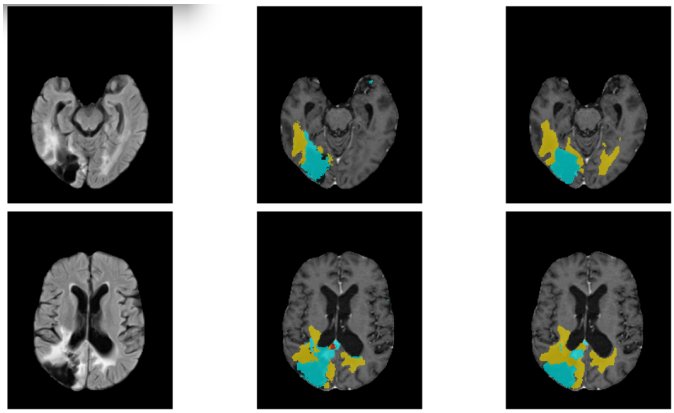

Fig. 1: High-performing test case BraTS-GLI-03056-101. Axial T2-FLAIR slices with GT (left) and predicted segmentation overlay (right). WT Dice = 0.8599, TC Dice = 0.7888, ET Dice = 0.7945.

B. Challenging Test Case

Patient BraTS-GLI-00511-101: WT = 0.5212, TC = 0.6105, ET = 0.6105. The model under-predicts edema (4,306 vs GT 15,672 voxels) and misses NCR subregion entirely.

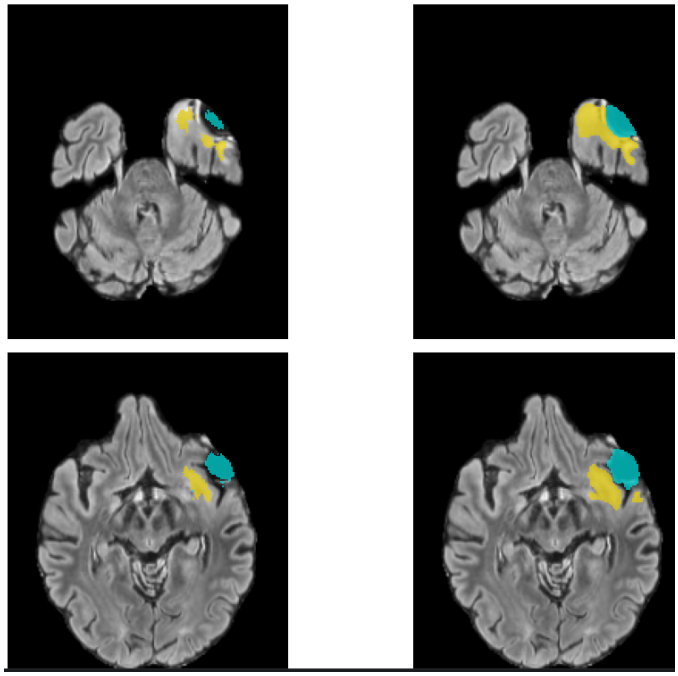

Fig. 2: Challenging test case BraTS-GLI-00511-101. Edema under-predicted heavily (4,306 vs GT 15,672 voxels) and missed NCR subregion. WT Dice = 0.5212.

C. Train vs. Test Dice Distribution (Box-Plots)

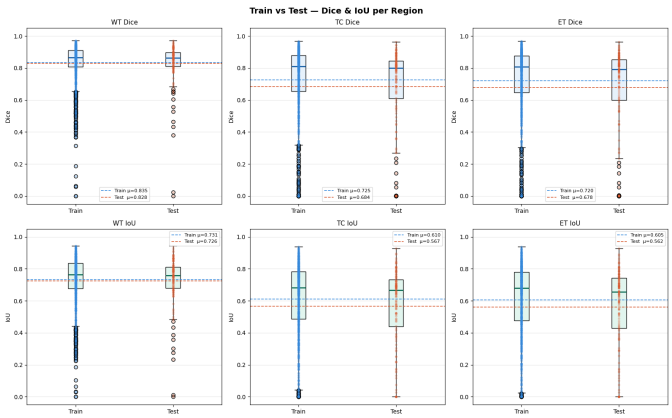

Fig. 3: Box-plot of Dice/IoU performance metrics for all three regions across 1,012 training (blue) patients in the training data set and 136 patients in the test data set (orange). Minimal distribution shift confirms no overfitting.

D. Mean Dice Bar Chart

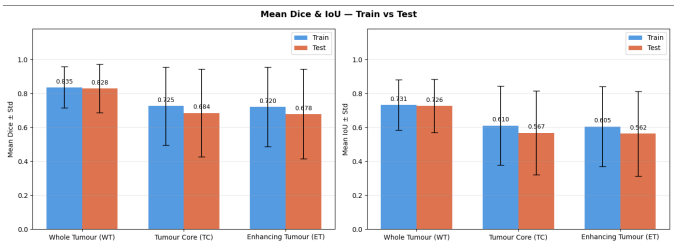

Fig. 4: Average Dice and IoU metrics for WT, TC, and ET on the training (blue) and test (orange) data sets. The vertical lines indicate one standard deviation.

E. Score Distribution Histograms

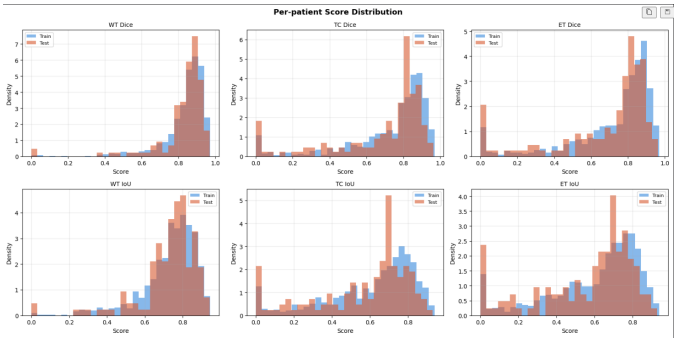

Fig. 5: Dice score histogram per patient. The left-skew tail indicates that most patients receive high Dice scores but there is a subset of patients with low Dice.

F. Per-Patient Scatter Plot

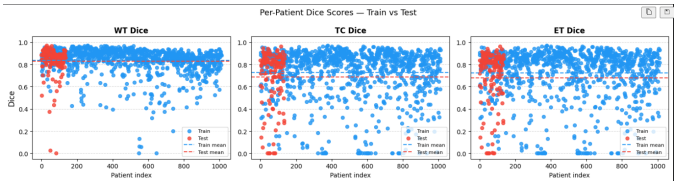

Fig. 6: Scatter plot of Dice scores by patients. Train-test clustering indicates no overfitting.

G. Threshold Tuning Heat-Map

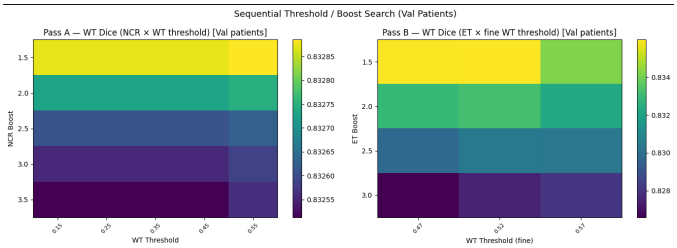

Fig. 7: Pass A: Tuning grid search for optimal threshold. Mean (WT+TC+ ET)/3 Dice score of 202 validation patients is shown via the colour gradient of the cells.The optimum (yellow) is robust to WT threshold variation at NCR boost of 1.5.

H. Additional Representative Test Cases

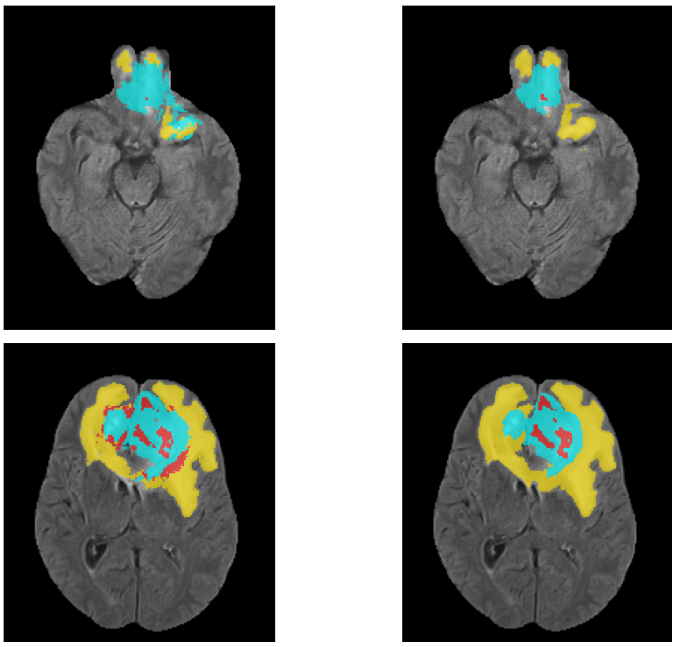

Fig. 8: Upper quartile test patient. All three subregions are accurately defined by precise, anatomically plausible boundaries.

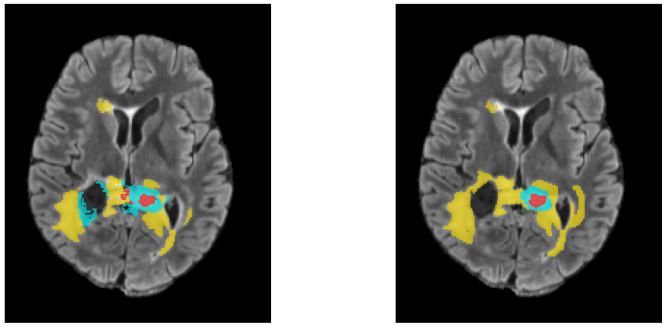

Fig. 9: Median-range test patient. Good segmentation of both tumour and ET, slight under-prediction of NCR is noted in tumour core region.

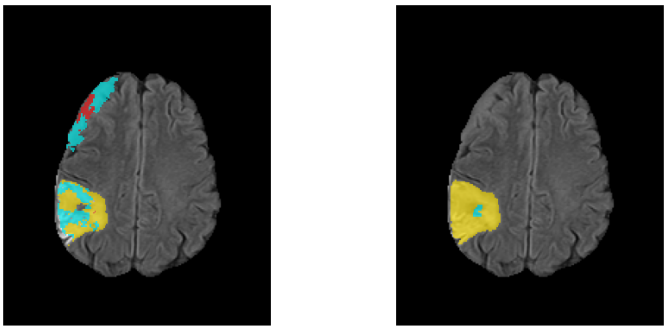

Fig. 10: Lower quartile test patient. Under-prediction of diffuse peritumoral edema region – the main limitation of the proposed pipeline.

X. Discussion

A. WT vs. TC/ET Gap

The ≈0.15 Dice gap between WT (0.8279) and TC/ET (≈0.68)can be attributed to three factors: (i) scarcity of NCR sub-region: — 0.11% of all brain voxels; Although NCR was 4× oversampled with weight 8.0, its performance was still low with F1 = 0.73; (ii) confusion between NCR and ET – wrong classification within tumour core region affects Dice scores of TC and ET but not of WT; (iii) lack of global context – voxel-wise independent prediction does not account for prior shape knowledge which can be used by CNN models.

B. MRF Effectiveness

Reduction of false positives – 429.6 blobs per patient on average. Compatibility matrix properly penalises ET–background border. Gaussian smoothing with $\sigma = 1$ might be too local to preserve large edema regions integrity.

C. Methodological Advantages

- **No deep learning:** completely interpretable and reproducible on any CPU.
- **Proper tuning:** thresholds determined based on validation samples only.
- **Proper regularization:** L1 + L2, subsampling, min-child-weight, γ to combat overfitting during all 500 iterations.
- **Scalablility** 500K voxels' batch processing can handle any sample without memory problems.

D. Disadvantages

- Independence of voxels: spatial context only appears in local statistics and Markov Random Fields.
- Cost of Feature extraction: about 45 sec for each patient.
- NCR performance: F1 = 0.73 is the weakest point in the pipeline.

XI. Future Work

1) **Cascaded approach**: First stage - Binary WT segmentation; second stage - Multi-class classifier in the WT mask. This eliminates the difficulty of sub-regions and simplifies the task to more balanced sub-space.
2) **Weighting function for focal loss:** $w_i = (1 - p_i)^\gamma$ where p_i corresponds to the predicted voxel probability, thus putting focus on hard cases (NCR/ET).
3) **Ring statistics:** statistics of the gradient of difference between T1c and T1n to help distinguish between rim and core.
4) **Dense CRF [5]:** bilateral kernel CRF considering both position and intensity of the voxel.
5) **Joint threshold optimization:** Bayesian optimization in the whole space of parameters (WT, NCR boost, ET boost)
6) **Shape priors:** PCA-based constraints on tumor shapes.

XII. Conclusion

With custom-built voxel features, XGBoost gradient boosting, and MRF spatial post-processing – without any deep learning –, the method delivers:

- **WT Dice 0.8279** surpasses the Phase 1 feasibility criterion of ≥0.80 and above.
- **TC Dice 0.6849 and ET Dice 0.6787** — novel subregion segmentation performance.
- **No overfitting**: whatsoever with a maximal difference of 0.0438 between the two methods in all six criteria.
- **I**mprovement of +10.3% WT Dice and **+15.6% WT IoU** compared to the Phase 1 binary benchmark.

The biggest open question is how to overcome the ≈0.15 Dice discrepancy between WT and TC/ET that stems from NCR infrequency and morphology.

References

[1] B. H. Menze *et al.*, "The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS)," *IEEE Trans. Med. Imaging*, vol. 34, no. 10, pp. 1993–2024, 2015.

[2] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proc. ACM SIGKDD*, pp. 785–794, 2016.

[3] C. Ma, G. Luo, and K. Wang, "Concatenated and Connected Random Forests with Multiscale Patch Driven Active Contour Model for Brain Tumor Segmentation," *IEEE Trans. Med. Imaging*, vol. 38, no. 8, 2019.

[4] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural Features for Image Classification," *IEEE Trans. Syst. Man Cybern.*, vol. SMC-3, no. 6, pp. 610–621, 1973.

[5] P. Krähenbühl and V. Koltun, “Efficient Inference in Fully Connected CRFs with Gaussian Edge Potentials,” in *NeurIPS*, vol. 24, 2011.
[6] S. Bakas *et al.*, “Advancing The Cancer Genome Atlas glioma MRI collections with expert segmentation labels,” *Nature Sci. Data*, vol. 4, p. 170117, 2017.
[7] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional Networks for Biomedical Image Segmentation,” in *MICCAI*, pp. 234–241, 2015.
[8] J. D. Lafferty, A. McCallum, and F. Pereira, “Conditional Random Fields,” in *Proc. ICML*, pp. 282–289, 2001.